Sparsity as a Lens for Interpretability: Discovering Functional Components in Vision Transformers

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Abstract

This minor thesis investigates the functional organization of Vision Transformers (ViTs) by discovering class-specific sparse subnetworks. We develop two approaches for discovering these subnetworks, showing that gradient-based pruning is more effective than evolutionary methods for learning performant sparse models. Our large-scale analysis across all 1000 ImageNet classes reveals a hierarchical structure: early layers contain general-purpose components used across most classes, while later layers exhibit increasing specialization. Remarkably, 43.1% of class-specific subnetworks achieve improved accuracy compared to the dense model while using only 8% of components, suggesting that pruning removes redundancy and interference for specific tasks. We identify specific attention heads with strong affinities for semantic categories, providing interpretable starting points for mechanistic analysis. These findings demonstrate that sparsity-based methods can serve as effective tools for understanding neural network internals and creating efficient, specialized models.

1. Introduction

Transformers have become the dominant architecture for machine learning models across a range of machine learning tasks. Their attention mechanism allows them to understand rich relationships between sequential data, most notably natural language [17]. This architecture has even been applied to computer vision, where Vision Transformers (ViTs) [5] have dominated vision benchmarks such as the ImageNet-1K challenge [14].

However, despite their empirical success, the inner workings of transformers remain largely a mystery. For ViTs, little is known about the functional organization of the model's components and how each contributes to the model's classification performance.

This minor thesis aims to fill this gap by investigating whether it is possible to identify *class-specific sparse subnetworks* within a pretrained ViT. By analyzing how class-specific accuracy varies as different components are pruned, we can uncover the relative importance of each component in the model's classification process, allowing us to better understand the inner workings of ViTs.

Concretely, this work addresses the following research questions:

RQ1 Are specific components of a ViT disproportionately responsible for the classification of certain image classes?

RQ2 What is the nature of the trade-off between model sparsity and classification accuracy when identifying a class-specific subnetwork?

RQ3 What are the most effective optimization strategies for discovering class-specific subnetworks?

Through the exploration of these research questions, this minor thesis makes the following contributions:

- A framework for identifying and evaluating class-specific sparse subnetworks in ViTs using gradient-based and multi-objective optimization.
- An analysis of the distribution of component usage across layers and classes in ViTs, revealing patterns of generalization and specialization.
- An analysis of the Pareto frontier between model sparsity and classification accuracy, which reveals the novel finding that pruning can in some cases increase class-specific accuracy.

The remainder of this thesis is structured as follows: Section 2 reviews related work on vision transformers, interpretability in transformers, sparsity, and subnetwork discovery. Section 3 introduces our proposed framework for identifying and evaluating class-specific sparse subnetworks. Section 4 presents the experimental setup and results that aim to answer the research questions. Section 5 interprets these results and, connecting them to our research questions and discusses their broader implications. Finally, Section 6 summarizes key findings and outlines directions for future work.

2. Related Work

This thesis sits at the intersection of three research areas: the architecture of ViTs, the interpretability of their internal components, and the discovery of sparse subnetworks.

Vision Transformers. The Vision Transformer (ViT) [5] is a transformer-based architecture adapted from the original Transformer [17] for computer vision. While the original Transformer was designed for natural language processing (NLP) which takes in tokenized sequential data, the ViT converts an image into a sequence by splitting a single image up into smaller patches. These patches are then embedded and used as input to the Transformer. The ViT's self-attention blocks allow it to have a global view of the image, contrasting to Convolutional Neural Networks (CNNs) that have local receptive fields which only allow them to see a small region of the image at a time. This architectural improvement has led to state-of-the-art performance on image classification benchmarks. However, the interactions between the ViT's attention and MLP blocks are complex, making its decision-making process opaque.

Interpretability in Transformers. Understanding the decision-making process of neural networks has long been a challenge. Many approaches in Explainable AI (XAI) have focused on post-hoc methods that use attribute features from the input to explain the model's decisions [8, 13, 15]. For Transformers, recent work in *mechanistic interpretability* has focused on reverse-engineering the model's internal computations into human-understandable processes [6, 10]. This research has shown that specific components, such as attention heads and MLP neurons, perform interpretable functions and algorithms [9, 11].

Most of the work in mechanistic interpretability has focused on NLP models, leaving a gap in the field for the exploration of component functionality in vision models.

Sparsity and Subnetwork Discovery. A well-established hypothesis in machine learning is the *Lottery Ticket Hypothesis* [7]. This hypothesis posits that large models are often over-parameterized; that there exists a smaller subnetwork within such a model that, when trained in isolation, can match the accuracy of the original model.

Subsequent work has developed techniques for identifying such subnetworks. Foundational work in this area found that many attention heads in Transformers are redundant and can be pruned with minimal impact on performance [18]. This principle has been adapted for ViTs, where similar findings have been made for the pruning of other components including MLP blocks and specific image patches [2, 12, 20]. The primary motivation for this line of research has been computational efficiency—reducing the cost of inference by removing redundant components. From an interpretability perspective, this reveals a missed opportunity for understanding the inner workings of these models through analysis of the pruned components, a gap that this thesis aims to bridge.

Summary and Research Gap In summary, prior work has established ViTs as a dominant architecture for computer vision, developed tools to probe their internals, and shown that they contain sparse, efficient subnetworks. However, a clear gap remains at the intersection of these fields. Few studies have used sparsity as a tool for interpretability to discover subnetworks that are causally linked to the classification of specific classes. By optimizing for sparsity and accuracy on a per-class basis, this thesis bridges this gap, using subnetwork discovery as a novel method for interpreting component functionality within ViTs.

3. Methodology

3.1. Problem Formulation

For this work, we consider the pretrained Vision Transformer Large model with a 16x16 patch size (ViT-L/16) [5], sourced from the Hugging Face Hub as `google/vit-large-patch16-224`. This model was fine-tuned on the ImageNet-1K dataset [14]. It comprises 24 transformer layers, each with 16 attention heads and 4096 hidden MLP units.

We measure class-specific accuracy as the proportion of correctly classified samples for a given target class c. Specifically, a prediction $\hat{y}$ given label y is considered correct in two cases:

1. $y = c \cap \hat{y} = y$, given a target class example, the model correctly predicts the target class.
2. $y \neq c \cap \hat{y} \neq c$, given a non-target class example, the model predicts any other class.

We denote the class-specific accuracy as accuracy^c as the proportion of correctly classified samples for a given target class' dataset.

The goal for this work is to identify how much we can prune a ViT's internal components while preserving its ability to correctly classify examples from a specific class.

3.2. Masking Strategies

Our pruning strategy involves learning a binary mask M^c for a specific class c that "zeroes out" the activations of selected components during the ViT's forward pass. To study the effects of sparsity at different structural levels, we experiment with three masking strategies—*layer-wise*, *block-wise*, and *unit-wise*—each corresponding to a different level of masking granularity.

Masking Strategy	Description	Mask Dimension (n)
Layer-wise	One mask per transformer layer	24
Block-wise	One mask per attention and MLP block	$24 \times 2 = 48$
Unit-wise	One mask per attention head and MLP neuron	$24 \times (16 + 4096) = 98688$

Table 1. Summary of masking strategies and corresponding mask sizes for the ViT-L/16 architecture.

Layer-wise masking. Layer-wise masking assigns a single binary variable to each transformer layer, controlling whether the output of that layer is retained or zeroed. This captures the relative importance of entire layers.

Block-wise masking. Block-wise masking assigns one binary variable to each attention and MLP block, producing 48 binary values in total. This allows the model to deactivate either the self-attention or MLP components independently across layers.

Unit-wise masking. Unit-wise masking assigns a binary variable to each attention head and neuron across all attention and MLP blocks, resulting in 98,688 mask entries. This provides maximum flexibility but dramatically increases the search space (2^n possible subnetworks).

The decision to mask along attention heads rather than individual neurons within those heads follows from the structure of the attention mechanism: each attention head output forms an interpretable subspace, whereas individual neuron activations within a head are not basis-aligned or semantically meaningful [6, 10].

3.3. Optimization Strategies

In addition to optimizing for accuracy^c, we also optimize for the sparsity^c of the mask defined as:

$$\text{sparsity}^c = 1 - \frac{\|M^c\|_0}{n},$$

where $\|M^c\|_0$ is the number of non-zero elements in M^c and n is the total number of maskable components.

We explored two complementary optimization strategies: a population-based multi-objective approach (NSGA-II) and a continuous relaxation approach optimized by gradient descent.

NSGA-II. The Non-dominated Sorting Genetic Algorithm II (NSGA-II) [4] is a multi-objective evolutionary algorithm that maintains a population of candidate solutions and iteratively refines them based on dominance ranking and diversity preservation. In this work, each candidate represents a binary mask $M^c \in \{0,1\}^n$, initialized randomly and evolved via crossover and bit-flip mutation. At each generation, candidates are evaluated using the accuracy^c and sparsity^c metrics, with non-dominated individuals carried over to the next generation for mutation.

NSGA-II was chosen for its ability to handle multiple objectives at once and its ease of implementation using the `pymoo` [1] framework. However, the discrete and evolutionary nature of NSGA-II makes it unscalable for exponentially large search spaces.

Gradient Descent. While NSGA-II is able to explore the discrete search space of the binary masks directly, gradient descent requires a continuous relaxation. For this, we learn a continuous proxy mask $\tilde{M}^c \in \mathbb{R}^n$ and map it to a pseudo-binary mask via sigmoid activations and a straight-through estimator (STE) [3, 19]:

$$M_i^c = \begin{cases} 1 & \text{if } \sigma(\tilde{M}_i^c) > 0.5, \\ 0 & \text{otherwise.} \end{cases}$$

During backpropagation, gradients are propagated through the sigmoid function, allowing $\tilde{M}$ to be updated using standard stochastic gradient descent.

A challenge with this approach is that accuracy^c and sparsity^c are not differentiable with respect to the binary mask M. Thus, for accuracy^c we instead use the binary cross-entropy loss $\mathcal{L}_{\text{BCE}}$ between the true labels y and the model's predictions $\hat{y}$. And for sparsity^c we compute with respect to the continuous-valued pseudo-binary mask $\sigma(\tilde{M})$ instead of M. Unlike NSGA-II which can handle multiple objectives at once, gradient descent requires a single continuous loss function:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(y, \hat{y}_M) + \lambda \, \|\sigma(\tilde{M})\|_1,$$

where λ controls the trade-off between classification accuracy and sparsity.

4. Experiments

This section details the empirical evaluation of our approach. We first describe the experimental setup (Section 4.1), then compare the different masking and optimization strategies to identify the most effective configuration (Section 4.2). Finally, we apply this configuration across all 1000 ImageNet classes and analyze the resulting subnetwork properties (Section 4.3).

4.1. Experimental Setup

For each of the 1000 classes in ImageNet-1K, we constructed a balanced binary classification dataset. Each

dataset contained all images of the target class (positive examples) and an equal number of randomly sampled images from the other 999 classes (negative examples). We then partitioned these datasets into training and validation sets using a 90/10 split. The same process was applied to the original ImageNet-1K validation set, which was held out as our final test set.

4.2. Comparison of Subnetwork Discovery Methods

We begin by comparing the NSGA-II and gradient descent optimization methods across the three masking strategies: layer-wise, block-wise, and unit-wise. Each experiment was run for 500 optimization iterations. For NSGA-II, this corresponded to 20 generations with a population of 25; for gradient descent, this was 500 update steps. For all experiments, masks were randomly initialized. Further details on all hyperparameters are provided in Appendix A.1.1.

To determine the optimal strategy for subnetwork discovery, we evaluated all six method combinations on a representative class, *Border collie* (ImageNet class 232). The resulting Pareto frontiers between class-specific accuracy and model sparsity, are shown in Figure 1 for each configuration.

The results reveal a clear interaction between the optimization algorithm and the granularity of the search space. As shown in Panels (a) and (b), NSGA-II is competitive with gradient descent where the search space is relatively small.

However, for unit-wise masking, Panel (c) shows gradient descent discovering a strictly dominant Pareto frontier. The exponentially larger search space renders NSGA-II's evolutionary approach ineffective, as its reliance on random bit-flip mutations is too inefficient for exploration given the budget of 500 iterations. On the other hand, the gradient-based method excels by leveraging a continuous relaxation of the problem. Results for a second class, *tabby cat* (ImageNet class 281), were consistent with these findings and are included in Figure 4.

This finding directly addresses RQ3: while evolutionary approaches like NSGA-II may be competitive in low-dimensional search spaces, gradient-based methods are superior in higher-dimensional spaces where all mask entries can be tuned at once. Thus, we adopt *unit-wise masking with gradient descent* for our subsequent experiments.

4.3. Large-Scale Analysis of Class-Specific Subnetworks

Having established our methodology, we applied it to all 1000 classes of the ImageNet-1K dataset. This large-scale analysis allows us to investigate our remaining research questions by examining the patterns in pruned subnetworks.

All hyperparameters were unchanged from Section 4.2, however, we used only 100 epochs with an early stopping of 10. Moreover, we used a fixed λ found from a hyperparameter sweep done in the previous experiment, shown in Appendix A.2. As most runs yielded several Pareto fronts, we used the Pareto frontier with the highest geometric mean between the test accuracy and sparsity:

$$M^{c*} = \arg\max_{M^c} \text{accuracy}^c \cdot \text{sparsity}^c,$$

4.3.1. Distribution of Component Usage: A Hierarchy of Specialization

To address RQ1—whether any components are disproportionately responsible for different classes—we analyzed the distribution of component usage across the model's 24 layers, averaged over all 1000 class-specific subnetworks. Section 4.3.1 plots the average percentage of retained components (attention heads and MLP neurons) in each layer.

The analysis reveals a distinct pattern of hierarchical feature learning. Component usage is highest in the early layers and declines steadily in later layers, indicating that initial layers perform more general tasks, whereas later layers perform tasks relevant to only a few classes. For attention heads usage remains high from the early to mid-layers with a steeper drop-off after layer 20. This suggests that for attention, specialized functionality is most prominent in the latest layers.

For MLP neurons, the usage rate is considerably lower than for attention heads, suggesting that they perform more specialized tasks. However, the sharp drop in MLP neuron usage after the very first layer suggests that in this initial layer, neurons are performing more general tasks, with the decreasing trend indicating that specialization becomes more prominent throughout the network. These findings align with the hypothesis that initial layers learn general, broadly applicable features (*e.g.*, edges, textures), which are essential for nearly all classes whereas later layers become progressively more specialized, with fewer components being relevant to any single class.

To address RQ1, we can identify which specific components have the highest and lowest usage rates, allowing us to identify both general-purpose and specialized components. For this experiment, we only consider attention heads, leaving the analysis of MLP neurons to future work. Table 2 lists the 12 attention heads with the highest and lowest average usage rate.

Unsurprisingly, heads in the early layers (1-6) have the highest usage rates reinforcing the idea that early-layer components perform critical and general-purpose functions. The head at layer 4 index 15 is of particular interest as it was used in 98.2% of classes. The computation this head performs is likely foundational for vision tasks, warranting further mechanistic interpretability.

On the other hand, most of the heads with the lowest usage rates are located in the last layer (24), again reinforcing the hypothesis that components in later layers are more

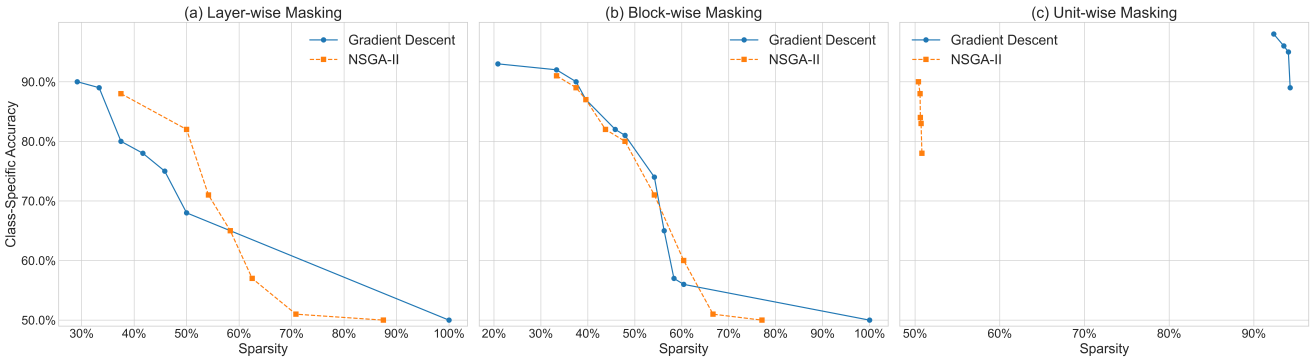

Figure 1. Comparison of Pareto frontiers for the Border collie class. Each panel compares Gradient Descent (solid line) and NSGA-II (dashed line) for a different masking strategy. (a) For layer-wise masking, both methods perform comparably. (b) For block-wise masking, performance is also comparable. (c) For unit-wise masking, gradient descent discovers a strictly dominant frontier, achieving a superior accuracy-sparsity trade-off compared to NSGA-II, which struggles in the high-dimensional search space.

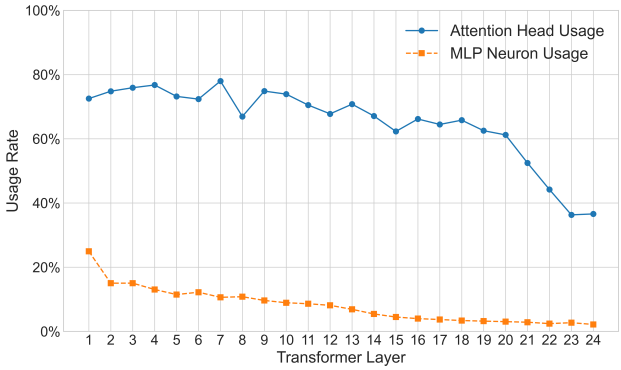

Figure 2. Average component usage per layer across all 1000 class-specific subnetworks. Early and middle layers are utilized more heavily, suggesting they capture more general-purpose features, while usage drops in the final layers, indicating specialization.

Layer	Head Index	Usage Rate (%)
4	15	98.20
1	8	96.70
3	4	96.50
5	8	95.70
6	2	95.60
3	7	95.10
⋮	⋮	⋮
8	6	19.50
24	10	19.40
24	8	18.30
24	3	17.10
24	13	15.90
24	4	13.50

Table 2. Top and bottom 6 most utilized attention heads across all 1000 ImageNet classes. The top heads represent the most general components in the ViT-L/16 model, while the bottom heads are the most specialized.

specialized. The attention head at layer 8 index 6 is an exception to this rule. To answer RQ1—are specific components disproportionately responsible for the classification of certain image classes—we analyzed the class themes associated with each of the six most specialized heads.

We categorized each of the 1000 ImageNet classes into the 11 superset categories curated by Tsipras *et al*. [16]. For each specialized head, we then calculated how frequently it was used for classes in each category C, relative to the baseline frequency of that category in ImageNet. This allows us to calculate the *affinity* of each head to each category:

$$\text{affinity}^C = \frac{\text{observed frequency}^C}{\text{expected frequency}^C},$$

where values greater than 1 indicate that the head is used disproportionately often for that category. The heads with category affinities > 1.5 are shown in Table 3.

Layer	Head Index	Dominant Categories	Affinities
24	13	Vehicles	2.68
24	3	Devices	1.60
24	4	Clothes	1.53
24	8	Other Mammals	1.61
8	6	Invertebrates	1.77

Table 3. Category affinity for the six most specialized attention heads. Only categories with enrichment scores > 1.5 are shown.

This analysis reveals that attention heads in later layers are indeed more specialized even at high-level categories.

The layer 24 index 13 head is of particular interest with an affinity of 2.68 for classes in the Vehicles category, suggesting that this component is responsible for the recognition of late-stage vehicle related features.

4.3.2. Analyzing the Sparsity-Performance Trade-off

Our second key finding addresses RQ2: the nature of the trade-off between sparsity and accuracy. After evaluating our 1000 class-specific subnetworks on the held-out test set, we discovered that pruning doesn't just maintain or slightly worsen accuracy, but can even improve it.

Metric	Value
Classes with Improved Accuracy	431 / 1000 (43.1%)
Average Accuracy Change	-0.30%
Average Subnetwork Sparsity	91.73%

Table 4. Summary statistics for subnetwork discovery across all 1000 ImageNet classes. Results show that near-baseline performance is achievable at extreme levels of sparsity.

As summarized in Section 4.3.2, our experiment found that, on average, subnetworks only used 8.27% of the base model's components while incurring only a 0.30% drop in accuracy. Surprisingly, 431 out of 1000 classes even saw an increase in accuracy. This suggests that for many tasks, the dense parent network contains components that introduce noise or interference for a given class. Pruning these components acts as a form of task-specific regularization that can improve performance.

The distribution of these performance changes is shown in Figure 3. To provide a qualitative understanding of the distribution's tails, Table 5 lists the classes with the largest accuracy gains and losses.

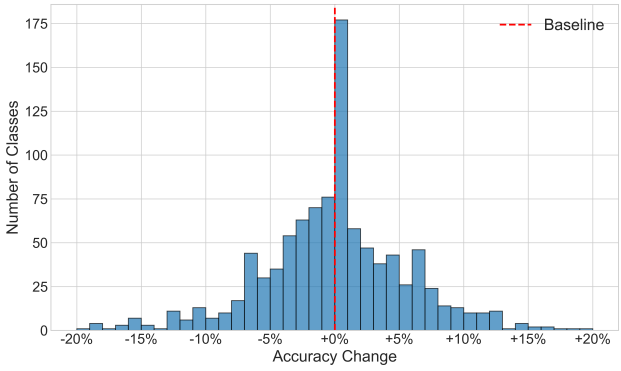

Figure 3. Distribution of accuracy changes across all 1000 ImageNet classes after subnetwork discovery. While the mean change is slightly negative, a significant portion of classes (43.1%) show improved accuracy, indicating that pruning can remove harmful interference for certain tasks.

Table 5. Top 10 and bottom 10 classes by accuracy improvement after subnetwork discovery.

Class Name	Sparsity	Baseline Acc.	Masked Acc.	Acc. Change
laptop, laptop computer	0.93	0.60	0.95	+0.35
Appenzeller	0.94	0.63	0.91	+0.28
maillot	0.93	0.60	0.81	+0.21
dock, dockage, docking facility	0.95	0.74	0.95	+0.21
tiger cat	0.93	0.64	0.84	+0.20
chiffonier, commode	0.96	0.75	0.93	+0.18
screen, CRT screen	0.92	0.65	0.82	+0.17
miniature poodle	0.93	0.74	0.91	+0.17
wolf spider, hunting spider	0.91	0.78	0.94	+0.16
cassette player	0.92	0.70	0.86	+0.16
⋮	⋮	⋮	⋮	⋮
padlock	0.94	0.91	0.65	-0.26
muzzle	0.91	0.88	0.61	-0.27
Australian terrier	0.96	0.82	0.53	-0.29
Greater Swiss Mountain dog	0.92	0.86	0.55	-0.31
tape player	0.92	0.82	0.50	-0.32
Labrador retriever	0.98	0.91	0.57	-0.34
scale, weighing machine	0.96	0.91	0.57	-0.34
mailbag, postbag	0.99	0.89	0.50	-0.39
Bernese mountain dog	0.92	0.96	0.56	-0.40
boxer	1.00	0.91	0.50	-0.41

From this table, we can see that classes that improve the most from pruning are often man-made objects with distinct features ('laptop', 'cassette player', 'commode'). Conversely, classes with subtle inter-class visual differences, such as various dog breeds ('boxer', 'Bernese mountain dog', 'Labrador retriever') seem to suffer the most from pruning. Notably, many of these worst-performing classes were pruned to extreme sparsity levels (>95%), suggesting that the drop in accuracy could be avoided with better settings for λ.

5. Discussion

This section summarizes the key findings from our experiments, connecting them back to our initial research questions. We then explore their broader implications and acknowledge the limitations of this work as well as directions for future research.

5.1. Interpretation of Key Findings

Our investigation had three key findings that provide insights into the functional organization of ViTs.

First, we found that the most effective subnetwork discovery strategy depends on the granularity of the search space and the corresponding choice of optimization method. For smaller search spaces, where we mask larger components of the model, both evolutionary and gradient-based methods performed comparably. However, for larger search spaces which are required for analyzing smaller components like attention heads and MLP neurons, evolutionary methods are no longer tractable. In this case the continuous relaxation of the gradient-based method is superior.

Second, our large-scale analysis revealed distinct patterns of generalization and specialization in ViTs. In early-to-mid layers, a larger proportion of components were re-

tained across all 1000 class-specific subnetworks, suggesting that these parts of the network perform general-purpose functions relevant to many classes. The declining usage rate of components in later layers indicates that components later in the network are more specialized, potentially dealing with more high-level or specific visual concepts.

Most significantly, our work challenges the conventional view of the relationship between model size and performance. We found that for 43.1% of classes, the pruned model outperformed the original dense model. This suggests that for many tasks, the dense parent network is not optimal, containing many redundant components that introduce noise or interference. Evidently, pruning these components can allow for a more specialized, task-aligned model with better performance.

5.2. Revisiting the Research Questions

RQ1: Are specific components of a ViT disproportionately responsible for the classification of certain image classes? Yes. Our analysis in Table 2 identified "general-purpose" attention heads in early layers, that are retained in over 95% of all class-specific subnetworks. Moreover, our affinity score analysis (Table 3) demonstrated that heads used by the least number of classes are also the most specialized. The head at layer 24 index 13 with an affinity of 2.68 for the "Vehicles" category is evidence that specific components are linked to the recognition and computation of high-level semantic concepts.

RQ2: What is the nature of the trade-off between model sparsity and classification accuracy when identifying a class-specific subnetwork? The trade-off is more complex than a simple Pareto frontier. While for many classes, increasing sparsity does lead to a decrease in accuracy, for a significant number of classes, the relationship is one of Pareto improvement where increasing sparsity leads to increased accuracy. This demonstrates that for certain tasks, the dense model is on a suboptimal part of the performance-computation curve, and that sparsity can result in superior, task-aligned subnetworks.

RQ3: What are the most effective optimization strategies for discovering class-specific subnetworks? The choice of optimization strategy depends on the granularity of the search space. As shown in Figure 1, for coarse masks, evolutionary and gradient-based methods are comparable. However, for the exponentially larger unit-wise search space, the gradient-based method is strictly dominant due to its ability to adjust all of its mask entries simultaneously in a smooth, continuous manner.

5.3. Broader Implications

The findings of this thesis have implications beyond the VIT-L/16 architecture, primarily in mechanistic interpretability and model efficiency.

From an interpretability perspective, this work reframes subnetwork discovery as a tool for causal analysis. We can use this methodology to find components that are necessary for completing a certain task, like with the "Vehicles" attention head. This can provide a starting point when deciding which components to further investigate using mechanistic interpretability techniques.

From an efficiency perspective, our results have practical applications. The ability to extract a subnetwork using only 8% of the original model's components while maintaining or improving performance allows for creating highly specialized, lightweight models for specific applications on low-resource devices.

5.4. Limitations and Future Work

Despite these contributions, this study has several limitations that provide directions for future work.

One primary limitation is the formulation of sparsity. Our current sparsity metric gives the same weight to MLP neurons and attention heads despite attention heads having significantly more parameters. This means that our unit-wise sparsity measures aren't an accurate reflection of the true parameter-level sparsity. Future work should investigate whether weighting the sparsity metric by the number of parameters provides different results in terms of component usage. Another key limitation is the use of a single, fixed hyperparameter (λ) for balancing accuracy and sparsity for the gradient-based approach. Our setting for λ was likely too high for certain classes, such as the dog breeds in Table 5 which resulted in suboptimal Pareto fronts. Developing a method for adaptive or per-class λ tuning may be a next step for pushing the Pareto frontier further.

This work also opens several avenues for future research.

Deepening Interpretability The generalized and specialized heads we identified are prime candidates for in-depth mechanistic analysis to understand the specific visual features they respond to and what types of computations they perform. Furthermore, the affinity analysis could be extended from high-level categories to more fine-grained sub-categories to uncover even more specific functional roles.

Extending to Language Models Applying this methodology to language models presents a challenging next step. Since there are no longer classes in language models, one would need to find another way to define linguistic tasks (*e.g.* syntax parsing, factual recall, *etc.*).

Analysis of MLP Neurons While a detailed analysis of MLP neurons was outside the scope of this work, conducting a similar analysis on the most and least utilized neurons could provide further insights into the model's structure.

Specialized Ensembles The collection of 1000 specialized subnetworks could be used to create an ensemble or a mixture-of-experts model, and its collective performance could be compared against the original dense model. This would answer the question of whether it's possible for specialized models to even outperform the dense model from which they were derived. This is an especially timely question given the current race to Artificial General Intelligence (AGI) which may be a misinformed goal.

6. Conclusion

This minor thesis set out to investigate the functional organization of Vision Transformers by identifying class-specific sparse subnetworks within a pre-trained Vision Transformer. By reframing network pruning as a tool for mechanistic interpretability, we have demonstrated that it is possible to identify small components that are necessary for the classification of specific semantic classes.

Our contributions are threefold. First, we developed a framework for finding task-specific sparse subnetworks using gradient-based and multi-objective optimization. Second, we provided a large-scale, quantitative analysis of the distribution of component usage across layers and classes and identified specific components of varying levels of generalization and specialization. Finally, we discovered that the trade-off between model sparsity and classification accuracy isn't just a Pareto frontier, but that increasing sparsity can in many cases improve performance. This finding challenges the idea that larger, general-purpose models are necessarily better and suggests that smaller, more specialized models can even outperform the dense model from which they were derived on specific tasks.

Ultimately, this work demonstrates that while the goals of interpretability and efficiency may seem different, they can both be used to complement each other. By optimizing for sparsity, we were able to find structure in ViTs. The discovery of these efficient, task-aligned subnetworks opens a promising new avenue for understanding the internal algorithms of Transformers and for building more specialized and efficient models in the future.

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A. Supplementary Experimental Details

This appendix provides additional details regarding the experimental setup, hyperparameter selection, and supplementary results referenced in the main body of the thesis.

A.1. Hyperparameter Settings

All experiments were conducted using a fixed random seed of 9 for reproducibility. The specific hyperparameters for the two main experimental phases are detailed below.

A.1.1. Initial Method Comparison (Section 4.2)

The following settings were used to generate the Pareto frontiers comparing NSGA-II and gradient descent for the *Border collie* and *tabby cat* classes.

NSGA-II

- Population Size: 25
- Number of Generations: 20
- Crossover Probability: 0.9
- Mutation Rate: 0.1
- Initial Sparsity: 0.5 (random initialization)

Gradient Descent

- Number of Epochs: 500
- Optimizer: Adam (`torch.optim.Adam` with default parameters)
- Batch Size: 32

A.1.2. Large-Scale Analysis (Section 4.3)

For the scaled experiment across all 1000 ImageNet classes, the gradient descent method was used with the following settings:

- Number of Epochs: 100
- Optimizer: Adam (`torch.optim.Adam` with default parameters)
- Early Stopping Patience: 10 epochs
- Sparsity Regularization (λ): 9.0
- Batch Size: 32

A.2. Selection of the Sparsity Regularization Parameter (λ)

To determine an effective value for the sparsity regularization parameter λ used in the large-scale analysis, we first performed a hyperparameter sweep during the initial method comparison experiments. This sweep explored different weightings for the two components of the loss function: the binary cross-entropy loss ($\mathcal{L}_{\text{BCE}}$) and the sparsity-inducing L1 norm ($\|\sigma(\tilde{M})\|_1$).

We tested five configurations of weights ($w_{\text{acc}}, w_{\text{sparsity}}$): (0.9, 0.1), (0.7, 0.3), (0.5, 0.5), (0.3, 0.7), and (0.1, 0.9). These were then used for the combined loss function:

$$\mathcal{L} = w_{\text{acc}} \cdot \mathcal{L}_{\text{BCE}} + w_{\text{sparsity}} \cdot \|\sigma(\tilde{M})\|_1$$

The configuration with $w_{\text{acc}} = 0.1$ and $w_{\text{sparsity}} = 0.9$ consistently produced superior Pareto frontiers for both the *Border collie* (class 232) and *tabby cat* (class 281) test cases. For the main paper's simplified loss function, $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \|\sigma(\tilde{M})\|_1$, this corresponds to a λ value derived from the ratio of the weights:

$$\lambda = \frac{w_{\text{sparsity}}}{w_{\text{acc}}} = \frac{0.9}{0.1} = 9$$

This value of $\lambda = 9$ was therefore fixed for all 1000 classes in the large-scale analysis to ensure consistency.

A.3. Supplementary Pareto Frontiers

To demonstrate that the findings presented in Section 4.2 were not unique to a single class, we replicated the experiment for the *tabby cat* class (ImageNet class 281). The results, shown in Figure 4, are consistent with those for the *Border collie* class, reinforcing that the gradient-based method with unit-wise masking is the most effective strategy for discovering high-performing sparse subnetworks.

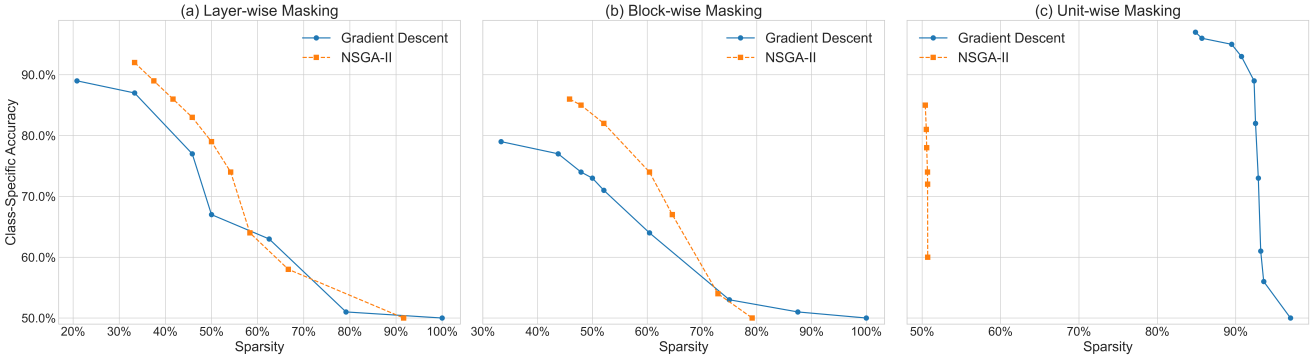

Figure 4. Comparison of Pareto frontiers for the Tabby cat class. The results mirror those in Figure 1, showing that gradient descent with unit-wise masking (c) discovers a strictly dominant frontier over NSGA-II.